



EMMANUEL MUGAMBI RIUNGU

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Skills

- Programming: Java | Spring Boot | Python | Django | React | TypeScript | Nextjs
- Tools: Git, Postman, MySQL, PostgreSQL, Kubernetes, Docker, Azure, Visual Studio Code, IntelliJ, Android Studio, DBeaver, MongoDB, Firebase, TensorFlow, Pandas, NumPy, Hugging Face Transformers, Gradio, LIME, Scikit-learn, PyTorch

Experience

1. eMobilis Technology Training Institute. Software Engineer Intern 01/2025 – 03/2025

- Developed cloud-based applications using Python, Django, and AWS to enhance system scalability.
- Integrated Firebase and PostgreSQL for real-time data storage in mobile applications.
- Implemented an AI-powered chatbot using NLP in collaboration with my team to improve customer interactions.
- Optimized API calls and database queries, reducing latency by 30%.

2. Safaricom PLC. Backend Engineer Intern 01/2026 – 06/2026

- Developed and supported RESTful API workflows using Java and Spring Boot within an enterprise systems environment.
- Tested and validated API requests, responses, and integration flows to support reliable communication between enterprise systems.
- Assisted in backend automation tasks aimed at reducing repetitive manual processes and improving operational consistency.
- Strengthened practical experience in Java, Spring Boot, REST APIs, enterprise integrations, backend automation, and system reliability.
- Participated in Agile sprint planning and code reviews.
- Collaborated with technical teams to review system changes, troubleshoot backend issues, and support production-facing system reliability.

Education

Bachelor of Science

STRATHMORE UNIVERSITY

Nairobi, Kenya

06/2022 – 12/2025

- Major in Informatics and Computer Science

Projects

• AgriScan: On-Device Plant Disease Detection Mobile App (Android-Kotlin, TFLite)

Trained a ResNet50 image classifier on 15 crop-disease classes, reaching 93.6% validation accuracy. Calibrated probabilities via temperature scaling with $T=0.5$ to reduce ECE from 0.148 to 0.011, then exported to Saved Model as TFLite and integrated into an Android app. Implemented on-device preprocessing and calibrated SoftMax in Kotlin via TFLite Interpreter with XNNPACK, returning top-k predictions plus uncertainty metrics such as entropy/quality to drive UX decisions. Find the project and its documentation [HERE](#).

- **Accounting and Inventory Management System (Django Framework)**

Contributed to the development of a web application integrating inventory and accounting modules for small eateries using Python and PostgreSQL to enable real-time tracking and efficient financial management. We achieved a streamlined system that empowers restaurant owners to manage resources effectively in a competitive industry, promoting sustainability. The application and its full documentation can be found [HERE](#).

- **Mental Health Detection System (DistilBERT, LIME)**

Built a Gradio-based NLP app using two fine-tuned DistilBERT models to detect mental health disorders and root causes from Reddit-like posts. Integrated LIME for explainable AI with HTML outputs. Enabled real-time analysis with confidence scores and interpretable predictions. The application and its full documentation can be found [HERE](#).

- **Java-Based Online Voting System (Spring Boot)**

Built an online voting system using Java, aimed at providing a seamless and secure voting experience, reducing the need for physical queues. We achieved enhanced accessibility and security in the voting process by reducing manual intervention and streamlining the voting procedure.

Certifications

- **Data Protection Certification** - Issued by Strathmore University
- **Kubernetes and Cloud Native Associate** – Issued by Linus Foundation [View Certificate](#)
- **Computer Networks (CCNA: Introduction to Networks)** - Issued by CISCO [View Badge](#)
- **Machine Learning Methods and Tools** - Issued by IBM [View Badge](#)
- **Data Analytics for Machine Learning** – Issued by IBM [View Badge](#)
- **Natural Language Processing** – Issued by IBM [View Badge](#)
- **Supervised Learning Methods** – Issued by IBM [View Badge](#)
- **Unsupervised Learning Methods** – Issued by IBM [View Badge](#)
- **Career Essentials in Generative AI** – Issued by Microsoft and LinkedIn [View Certificate](#)
- **Artificial Intelligence Fundamentals** - Issued by IBM [View Certificate](#)